

Arohan Ajit

San Jose, CA • arohan.ajit@outlook.com • 276-759-3566 • github.com/arohanajit • linkedin.com/in/arohanajit •

arohanajit.github.io

SUMMARY

Full Stack Engineer with 3+ years of experience building scalable integration solutions and enterprise applications. Proven track record in API design, system optimization, and cross-functional collaboration. Passionate about innovative solutions and business process automation through technology

TECHNICAL SKILLS

Languages: Python, Javascript, Golang, SQL • **Frontend:** React, Redux, TypeScript, Bootstrap, Tailwind CSS • **Backend:** Node.js, Django, Flask, REST APIs • **Database:** PostgreSQL, MongoDB, MySQL • **Tools & Technologies:** Docker, Kubernetes, AWS/GCP, GitHub Actions

EXPERIENCE

Software Development Intern, Chirpn, San Jose Jun 2024 – Nov 2024

- Designed and implemented scalable integration solutions using Node.js/Express, improving system performance by 30%
- Integrated React frontend with Python/Node.js backends, improving UI responsiveness by 20%.
- Automated CI/CD pipelines with GitHub Actions and Docker, reducing deployment effort by 40%.
- Architected scalable distributed systems using microservices pattern
- Managed multi-environment cloud infrastructure using AWS/GCP, cutting costs by 15%.

Software Development Associate, Accenture, India May 2021 – Jul 2023

- Built enterprise-grade web applications with Django / React that serve 10k + monthly users at 99.9% uptime.
- Improved query performance by 15% using Django ORM and PostgreSQL schema optimizations.
- Utilized Django's ORM and REST framework to build scalable and secure backend systems

Graduate Research Assistant, Center of Geospatial Analysis, NC State, Raleigh Sep 2024 – Present

- Improved raster data storage efficiency by 25% using quantization algorithms, reducing cloud storage costs for large-scale datasets on one of the major open source geospatial analysis softwares

PROJECTS

Distributed Key-Value Store | Golang, REST API, Consistent Hashing

Implemented a distributed key-value store featuring consistent hashing for data distribution, configurable replication factors, and fault tolerance. Built a thread-safe storage layer with content type preservation and shard management. Developed comprehensive REST APIs with middleware for request routing and timeout handling. Implemented cluster management for node discovery and health checking. Created extensive test suites covering unit tests, concurrent operations, and system stress testing.

Distributed Container Platform | Kubernetes, Docker, Python, React, ClamAV, Prometheus

Architected a production-grade malware scanning service using microservices architecture with Kubernetes orchestration. Implemented comprehensive monitoring using Prometheus, maintaining 99.9% system availability through Kubernetes probes and achieving sub-2 second response times for file analysis. Developed core scanning service in Python using microservices architecture. Implemented distributed processing system handling concurrent file analysis

GO Microservices Architecture Framework | Golang, Docker, PostgreSQL, Consul, Gin

Built a distributed e-commerce system with 6 microservices (user, product, order, payment, inventory management) implementing service discovery, API gateway routing, and containerized deployment. Implemented distributed consensus for service coordination. Integrated health monitoring, JWT authentication, and database migrations while ensuring inter-service communication using REST APIs and Consul service registry. Orchestrated containerized services using Kubernetes.

EDUCATION

North Carolina State University, MS in Computer Science Aug 2023 - May 2025

GPA: 3.8/4.0 | Coursework: Algorithms, Cloud Computing, Neural Networks, Computer Networks, Security

Kalinga Institute of Industrial Technology, BTech in Information Technology Aug 2017 - Aug 2021

GPA: 3.5/4.0 | Coursework: Operating Systems, Data Structures, OOP, Computer Networks

ACHIEVEMENTS

Microsoft Certified: Azure Fundamentals | Azure AI Fundamentals | IBM Professional Certificate